THE CAREER EXPERIMENT

Product discovery progress report

A plain-language record of what has been built, what went wrong, how the prompts corrected it, and how the current prototype works behind the screen.

Checkpoint saved locally

Repository: D:\Documents\ChatGPT\The Career Experiment Lab
Branch: codex/evidence-flow-refinements
Code checkpoint: f97bd3e
Report date: 18 August 2026

Current prototype promise

Use past work as evidence, compare how that work transfers to two possible roles, identify what remains uncertain, and run supported work trials that add new evidence without declaring a career winner.

This document is a handoff, not a production specification. It records the current product-discovery state so future work can continue without losing the reasoning behind the prototype.

1. Executive snapshot

The prototype has grown from a short onboarding mock-up into a connected evidence journey.

Area	What works now
Career comparison	Choose exactly two peer-level roles and compare the same activity across both.
CV foundation	Local PDF and Word text extraction, structured experiences, manual correction, and no permanent CV storage.
Past evidence	Activities are separated, normalised, deduplicated, linked back to source experiences, and reviewed individually.
Evidence map	Role importance, user preference, and self-confidence remain separate; unknown areas remain visible.
Career experiments	Supported work trials use a primer, first attempt, qualitative review, focused revision, preference, reaction, and synthesis.
Safety	No career-fit percentage and no automatic career recommendation.

Current verification

51 domain tests pass. ESLint, TypeScript checking, and the production build pass. The normal development app is available at http://localhost:3000/ while the server is running.

- The code is saved in Git on the local D-drive repository.
- The API secret remains in .env.local and is ignored by Git.
- The latest code checkpoint has not been pushed to GitHub automatically.

2. The current user journey

The product now follows one evidence chain instead of treating the CV, ratings, and experiment as separate features.

Step	Screen	Purpose
1	Choose roles	Select exactly two roles to compare.
2	Upload CV	Extract text locally from PDF or Word.
3	Choose experiences	Select the roles and projects that should influence this decision.
4	Review activities	Edit, remove, or add separate work activities.
5	Evidence tunnel	Review at most ten useful transferable activities.
6	Starting evidence	See what transfers, what the user wants, self-confidence, and role-specific unknowns.
7	Supported work trial	Choose a question, complete realistic work, use feedback, revise, and add new evidence.

The conceptual Career Evidence Matrix

For each activity, the application keeps four questions separate:

Past evidence	Preference	Role A	Role B
Have I done it? Across which experiences?	Do I want more, the same, or less?	How important is it, and how is it used?	How important is it, and how is it used differently?

Core product rule

Past experience is evidence that work happened. It is not proof that the user enjoyed it, performed it objectively well, or should choose the career that contains it.

3. What was built over time

Milestone	Main addition	Why it mattered
Foundation	Next.js + TypeScript prototype, seven-step local flow, AGENTS.md product guardrails.	Made the discovery hypothesis testable without production infrastructure.
CV evidence	PDF/Word upload, structured experiences, source provenance, manual experience and activity editing.	Replaced one manually typed job with a broader view of the user's history.
Transfer mapping	Career activity profiles, canonical activities, semantic transfer rules, maximum-ten tunnel.	Moved beyond literal keyword matching toward transferable work.
Evidence interpretation	Preference and confidence kept separate, role-specific evidence maps, neutral unknowns.	Avoided turning partial evidence into a recommendation.
Supported trials	Shared scenarios, role primers, first attempt, AI rubric review, focused revision, reflection, synthesis.	Made experiments feel like supported work rather than a quiz.
Generalisation	Local trial definitions for all prototype roles and three scenario families.	Allowed future comparison beyond the first PM vs behavioural case.

Version-control checkpoints

- 180e165 - Initial commit
- a849265 - Build Career Experiment discovery prototype
- 5a4233d - Refine career evidence discovery flow
- f97bd3e - Checkpoint career evidence and experiment prototype

4A. Mistakes, prompts, and corrections

The most useful progress came from dogfooding. The user repeatedly identified where the implementation matched code structure rather than human meaning.

What went wrong	How the user redirected it	What changed
One whole job description became one activity.	One experience can contain many separate activities; add repeatable fields and multi-line paste.	Experience and Activity became separate models. Manual and CV paths now converge into the same downstream activity pipeline.
PDF text was flattened into a page-long string.	Fix the PDF-to-text foundation before patching more experience regular expressions.	PDF items are reconstructed into approximate lines using their vertical and horizontal positions.
Some dates, awards, or programme text became job titles.	Keep title, organisation, dates, type, and description separate; do not display malformed parsing.	A stricter Experience contract and filtering reduced invalid experience cards.
Later CV roles had no activities.	Do not stop after the first roles; preserve and extract all meaningful work entries.	Experience detection and description boundaries were widened; manual correction remains available when source structure is ambiguous.
Edits made after returning to a prior screen did not reach the evidence flow.	Newly entered tasks must register when the user proceeds again.	Derived normalised and top-evidence state is recalculated from the edited experiences instead of retaining stale results.

What we learned

Extraction has two different jobs: first recover readable source text; then interpret that text into experiences and activities. Treating them as one problem made failures hard to diagnose.

4B. Mistakes, prompts, and corrections - meaning and evidence

What went wrong	User's correction	Current response
Literal keywords decided transferability.	Qualitative research is research; proposals involve analysis and recommendations; liaison work is stakeholder communication.	A canonical activity catalogue now uses phrases, capabilities, and role profiles rather than exact labels alone.
Long resume sentences appeared as activity titles.	Rephrase the evidence into a concise transferable skill and keep the original sentence only as evidence.	Canonical labels are shown in the main UI; original wording is retained behind provenance.
Useful work was labelled Unknown for both roles.	Find the closest responsibly supported transferable activity; it only needs to matter to one selected role.	Fallback mappings cover recurring work such as onboarding design, programme implementation, journey design, launch planning, and process optimisation.
Activity categories were redundant and sometimes wrong.	Do not make the user choose both a section and a duplicate category; use clearer groups such as Execution or Product work.	Category is derived from the canonical activity and is editable where necessary; unmatched work remains explicit rather than silently invented.
Role descriptions were repetitive boilerplate.	Explain the actual core task inside the activity and how it applies differently to each role, in no more than two sentences.	Role profiles supply concise, activity-specific relevance text outside the React components.
Quantitative analysis looked unimportant for PM.	PMs use data for product performance, user behaviour, experiments, and decisions.	The PM career profile now treats quantitative/data-informed analysis as meaningful role evidence.

4C. Mistakes, prompts, and corrections - interpretation and experiments

What went wrong	User's correction	Current response
The Top-3 screen repeated information already collected.	Use preference and confidence to organise evidence; avoid asking the same question twice.	Priority selection was simplified and the evidence map orders work by role importance, then preference and confidence.
Evidence cards had too many tags.	Keep the order but explain it in plain text rather than repeating three badges on every line.	The evidence map uses a concise activity list with fewer visual labels.
The first experiment was answer -> preference -> shallow recap.	Teach the role's way of thinking, evaluate, coach one area, then observe revision and reaction.	The supported trial separates first response, qualitative performance evidence, feedback, revision, learning response, preference, and work reaction.
Low knowledge looked like low ability.	Missing knowledge is not proof of low potential or poor fit.	The evaluator classifies knowledge, reasoning, communication, and insufficient-evidence gaps separately.
AI review content rendered as a wall of words.	Make the feedback readable and restore the scenario context during focused revision.	Review criteria are displayed as structured cards; revision includes the relevant brief and question.
Optional reflection was merely quoted back.	Infer a simple, useful next action; do not repeat vague disclaimers.	Reflection interpretation now produces plain-language context and a small action the user can try.
Try another experiment repeated LearnLoop.	Show another question or end honestly; use the same company/industry while changing the nature of the work.	A local scenario catalogue now includes LearnLoop plus additional same-context role trials.

5. How it works behind the screen

The application is intentionally local and modular. Each layer has one job so parts can be replaced without rebuilding the interface.

Layer	Responsibility	Key locations
UI and flow	Chooses the active screen and renders reusable consumer-style controls.	app/page.tsx; components/screens/*; app/globals.css
CV text	Reads PDF or Word in the browser and preserves readable text structure.	lib/cv/*; lib/pdf/*; lib/word/*
Experience extraction	Turns source text into structured roles and activity records.	lib/extraction/*
Activity meaning	Normalises wording, deduplicates work, preserves source IDs, and maps transferable components.	data/activityCatalog.ts; lib/evidence/normalizeActivities.ts; semanticActivityMapping.ts
Career knowledge	Defines each role's canonical activities, importance, and relevance descriptions.	data/careers.ts
Evidence logic	Chooses up to ten activities and builds role-specific existing evidence, preferences, confidence, and gaps.	lib/evidence/*
Work trials	Stores scenarios, role tasks, primers, rubrics, state transitions, and reflection interpretation.	data/experiments.ts; data/roleTrials/*; lib/experiments/*
AI evaluator	Sends only the trial brief and answers to a server-side endpoint and validates structured results.	app/api/evaluate-experiment/route.ts; lib/experiments/evaluationSchema.ts

Data flow

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CV file -> browser text extraction -> structured experiences -> selected experiences ->
individual activity records -> canonical activities with source IDs -> top evidence activities
-> separate preference and confidence answers -> role-specific starting evidence -> selected
uncertainty -> supported work trial.
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6. The important data distinctions

Concept	What it stores	What it must not imply
Experience	Role/project title, organisation, optional dates/type/description, and many activities.	A job is not one giant activity.
Activity evidence	Concise canonical label plus original wording and source experience IDs.	Doing the activity does not mean enjoying it.
Preference	More, about the same, or less.	Preference is not capability.
Skill confidence	Low, medium, or high self-assessment.	It is not objective performance evidence.
Career relevance	Core, Important, Supporting, Limited, or responsibly Unknown, plus a role-specific explanation.	Importance is not a fit score.
Experiment performance	Qualitative rubric observations about one response.	A simulation is not sustained job performance.
Learning response	Whether a focused revision visibly used feedback.	One revision is not a learning-potential score.
Work reaction	How the supported activity felt after instruction and feedback.	A reaction to one task is not a career verdict.

Why this structure matters

The product can later compare self-confidence with observed experiment evidence without overwriting either. It can also say that an unfamiliar method confounded preference instead of calling it poor performance.

7. Example: turning a CV sentence into useful evidence

The latest correction demonstrates the intended semantic mapping pattern.

Original CV evidence

Full sentence retained as provenance

Enhanced case-processing efficiency by 30% across 2,000+ cases and tripled reporting need by optimising academic integrity audit and reporting workflows.

What the user should review

Canonical activity	Transferable meaning	Provenance
Process optimisation	Improving a workflow to increase throughput, reliability, or efficiency.	Academic Integrity Operation Assistant - University Canada West

- The outcome numbers support the evidence but do not become the activity title.
- The phrase 'optimising ... workflows' provides a strong semantic signal for process optimisation.
- The canonical ID allows equivalent wording from another role to merge without losing either source experience.
- Role mapping then asks how process optimisation matters to each selected career; it does not rely on the exact words in the CV.

Generalisation strategy

For other resumes, the deterministic catalogue recognises action-plus-object patterns and outcome-led sentences. Examples include improving a process, redesigning a workflow, reducing turnaround time, streamlining reporting, or increasing operational efficiency. Ambiguous evidence still remains editable rather than being silently over-interpreted.

8. Supported career work trials

The experiment is designed to reduce obvious knowledge barriers before interpreting the user's reaction.

Stage	Evidence created
Choose a question	The uncertainty the user wants to investigate; the app does not choose a career.
Shared scenario	The same company and industry context for both roles, reducing domain-preference confounding.
Role primer	A short mental model that makes unfamiliar work accessible.
First attempt	The user's original reasoning is preserved.
Qualitative review	Rubric observations with Strong / Developing / Emerging / Not enough evidence.
Targeted feedback	One development opportunity plus a useful idea, not a model answer.
Focused revision	A separate answer showing how feedback was used.
Re-evaluation	Clear / some / little visible improvement, or not enough evidence.
Preference and reaction	How the work felt after support; stored separately from performance.
Synthesis	What became clearer, what looks developmental or confounded, and what remains unknown.

AI's narrow role

AI is used only to evaluate and coach free-text work-trial responses. It is not a chatbot, does not parse the CV in the current prototype, and does not recommend a career.

9. Career and experiment coverage

Prototype role catalogue

- Product Manager
- Behavioural Science Consultant
- Data Scientist
- Product Analyst
- UX Researcher
- Management Consultant
- Consumer Insights Researcher

Local scenario families

Scenario	Stable context	What changes between roles
LearnLoop	University study-planning product with week-two retention problems.	Product decisions, behavioural mechanisms, analytics, research, consulting, or insight work.
LearnLoop Recovery	The same learning-product domain and related engagement problem.	A second focused uncertainty rather than replaying the identical questions.
CampusPath	A university-facing product/service context.	Role-specific outputs and reasoning while the company and problem remain fixed.

Trial definitions are local mock data. The architecture supports different role lenses, but this is still a product-discovery catalogue rather than a validated library of occupational simulations.

10. Repository map for a beginner

If you want to change...	Start here
The order of screens and local state	app/page.tsx
The visual appearance	app/globals.css
A particular screen	components/screens/
Career titles, role activities, importance, and role descriptions	data/careers.ts
Transferable activity names, signals, components, and categories	data/activityCatalog.ts
How activities merge and preserve source experiences	lib/evidence/normalizeActivities.ts
How a work activity is mapped to both roles	lib/evidence/semanticActivityMapping.ts
Which maximum ten activities reach the tunnel	lib/evidence/selectTopEvidenceActivities.ts
How existing evidence and unknowns are assembled	lib/evidence/buildStartingEvidence.ts
PDF, Word, and CV parsing	lib/pdf/, lib/word/, lib/cv/, lib/extraction/
Experiment scenarios and role tasks	data/experiments.ts and data/roleTrials/
Experiment state and AI evaluation	lib/experiments/ and app/api/evaluate-experiment/route.ts
Regression checks	tests/domain.test.mjs

Local data and privacy

- The uploaded CV and extracted text live only in the current browser session.
- There is no account, database, or permanent CV storage.
- The OpenAI key belongs only in .env.local and never in client-side code or Git.
- The evaluation request sends trial context and answers through a server-side route; normal onboarding state stays local.

11. What remains imperfect

The prototype is substantially more coherent, but deterministic extraction and mapping still have honest limits.

Limitation	Current safeguard	Likely future replacement
PDF visual layouts vary widely.	Approximate positional line reconstruction plus manual correction.	Better document-layout parsing and broader fixture testing.
Experience detection can miss or mis-bound unusual CV sections.	Strict cards, no raw malformed title display, and Correct details.	Validated AI-assisted structured extraction with source citations.
Deterministic activity mapping cannot understand every sentence.	General action/object patterns, canonical catalogue, transparent Unknown state, and editable wording.	AI semantic mapping constrained to the career/activity ontology and validated against source text.
Career profiles are expert-informed mock definitions.	Descriptions are local, visible, and easy to revise.	Research with practitioners, occupational sources, and task-frequency evidence.
AI evaluation may be inconsistent or unavailable.	Validated schema, retry state, answers preserved, qualitative language, no fit score.	Model evals, prompt/version tracking, benchmark responses, and human review.
One short work trial has low external validity.	Cautious synthesis and explicit remaining unknowns.	Multiple scenarios and higher-fidelity field experiments.

Important interpretation boundary

Low performance may reflect missing knowledge, unfamiliarity, communication, or weak reasoning in one response. It must never automatically mean the user should not pursue that career.

12. Prompting lessons from this build

The most effective prompts named the broken evidence relationship, supplied a concrete example, and stated what must remain separate.

Useful prompt pattern	Why it worked
State the incorrect relationship	Example: 'one experience is becoming one task.' This identifies the data-model error, not just the visual symptom.
Show before and after	Example: a full sentence should become 'Process optimisation' while retaining the sentence as provenance.
Name non-negotiable distinctions	Preference is not capability; missing knowledge is not low ability; career importance is not a recommendation.
Use a realistic failure example	Examples such as liaison with government, creating proposals, or optimising workflows expose brittle literal matching.
Constrain the scope	Requests such as 'do not rebuild' and 'do not add the next feature' kept changes reversible.
Ask for verification	Specific manual cases plus lint, type, build, and regression tests made fixes durable.

A strong future prompt

Reusable structure

Explain the product decision being tested; show one or two concrete failures; define the desired data relationship; say what must not be inferred; identify the screens and layers allowed to change; and provide an acceptance example that can become a regression test.

13. Checkpoint and handoff

Saved state

The working implementation is committed locally on branch codex/evidence-flow-refinements at f97bd3e. This PDF is stored inside output/pdf/ in the same repository and can be committed as a separate documentation checkpoint.

How to continue safely

- Open the repository at D:\Documents\ChatGPT\The Career Experiment Lab.
- Read AGENTS.md before making changes.
- Run the normal development app and reproduce the user-facing issue before editing.
- Change the narrowest data or logic layer that owns the problem.
- Add a regression test for every semantic mapping or state-flow bug that is fixed.
- Run domain tests, ESLint, TypeScript checks, and a production build before committing.
- Never commit .env.local or paste an API key into chat or screenshots.

Recommended next discovery questions

- Can independent users understand the difference between past evidence, self-belief, and observed experiment evidence?
- Do the maximum-ten activities feel selective enough without hiding important work?
- Which semantic mappings are genuinely wrong across varied resumes rather than only missing a synonym?
- Does a second scenario change the user's reaction, or merely repeat the first result?
- Which role descriptions need validation with practitioners before the catalogue is trusted?

End of report

The prototype is still a discovery tool. Its strongest progress is not that it can label a career, but that it now keeps different kinds of evidence separate and makes uncertainty visible enough to test.